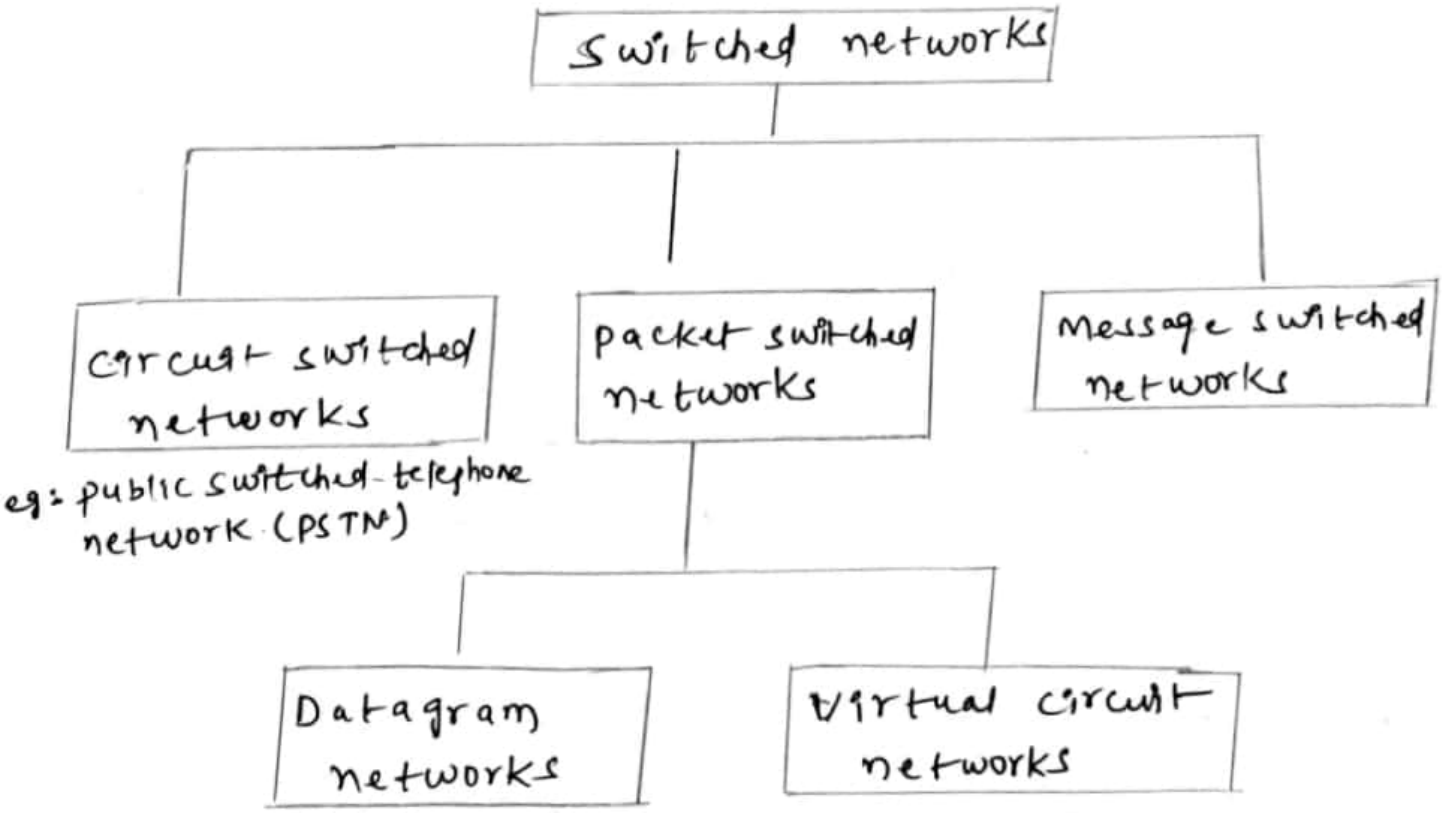
A switched network consists of a series of interlinked nodes, called switches. switches are devices capable of creating ~~creating~~ tempory connections b/w two or more devices linked to the switch.

→ Traditionally 3 methods of switching have been important. 1) circuit switching 2) packet switching 3) message switching.

→ we can divide today's n/w's into 3 broad categories.

Taxonomy of switched networks



Scanned with CamScanner

Circuit switching : circuit switching takes place at physical layer.

communication via circuit switching implies that there is a dedicated communication path between two stations. circuit switching uses either of two technologies: the space division switch or time division switch.

→ The path is a connected sequence of links between network nodes.

→ on each physical link, a logical channel is dedicated to the connection. In circuit switching, the resources need to be reserved during the setup phase, the resources remain dedicated for entire duration of data transfer phase until teardown phase.

There are 3 important phases in circuit switching.

1) circuit establishment (or) setup phase
2) Data Transfer
3) circuit disconnect (or) teardown phase.

Circuit establishment

→ To establish an end-to-end connection before any transfer of data.

→ Some segments of the circuit may be a dedicated link, while some other segments may be shared.

Data Transfer

→ Transfer data is from source to destination

→ The data may be analog or digital, depending on the nature of the network.

→ The connection is generally full duplex.

Circuit disconnect :-

→ Terminate connection at the end of data transfer.

→ signals must be propagated to deallocate the dedicate the dedicated resources.

Scanned with CamScanner

Advantages:

1) Fixed bandwidth, guaranteed capacity (no conge

2) Low-variance in end-to-end delay (constant delay).

Disadvantages:

1) circuit establishment and circuit disconnect introduces extra overhead and delay.

2) constant data rate from source to destination.

3) channel capacity is dedicated for the duration of the connection, even if no data is being transferred.

4) other users cannot use it even if free of traffic.

5) Not suitable for bursty data.

Packet switching:

→ In packet switching, there is no resource allocation for a packet. This means that there is no reserved bandwidth on the links, and there is no scheduled processing time for each packet. Resources are allocated on demand.

→ Data are transmitted in short packets (few kbytes).

→ A longer message is broken up into a series of packets.

→ Every packet contains some control information in its header (required for routing and other purposes).

→ Two approaches are commonly used for handling these packets: 1) virtual circuit 2) Datagram.

→ In packet switched network, there is no resource reservation; resources are allocated on demand.

Scanned with CamScanner

tual circuit Networks

→ A preplanned route is established before any packets are sent.

→ call Request and call Accept packets are used to establish the connection.

→ Route is fixed for the duration of the logical connection (like circuit switching).

→ Each packet contains a virtual circuit identifier as well as data.

→ Each node on the route knows where to forward packets.

→ A clear Request packet issued by one of the two stations terminates the connection.

→ Route between stations is set up prior to data transfer.

→ A packet is buffered at each node, and queued for output over a line.

→ A data packet needs to ~~needs~~ carry only the virtual circuit identifier for effecting routing decisions.

→ Intermediate nodes take no routing decisions.

→ often provides sequencing and error control.

→ In virtual circuit Network three phases will be there. In setup phase the source and destination use their global addresses to help switches make table entries for the connection.

→ In teardown phase, the source and destination inform the switches to delete corresponding entry.

→ Data transfer occurs b/w these two phases.

Scanned with CamScanner

Datagram approach

→ In this approach ther is no setup or teardown phases.

→ Each packet is treated independently, with no reference to packets that have gone before.

→ Every intermediate node has to take routing decisions.

→ Every packet contains source and destination addresses.

→ Intermediate nodes maintains routing tables.

Advantages :

→ call setup phase is avoided. (for transmission of a few packets, datagram will be ~~transf~~ faster).

→ Because it is more primitive, it is more flexible.

→ congestion / failed link can be avoided (more reliable).

Problems :

→ packets may be delivered out of order.

→ If a node crashes momentarily, all of its queued packets are lost.

Scanned with CamScanner

Comparison

Circuit Switching	Datagram Packet	Virtual Circuit Packet
Dedicated path	No dedicated path	No dedicated path
Path established for entire conversation	Route established for each packet	Route established for entire conversation
Call set up delay	Packet transmission delay	Call set up delay, Packet transmission delay
Overload may block call set up	Overload increases packet delay	Overload may block call set up and increases packet delay
No speed or code conversion	Speed or code conversion	Speed or code conversion
Fixed bandwidth	Dynamic bandwidth	Dynamic bandwidth
No overhead bits after call set up	Overhead bits in each packet	Overhead bits in each packet

Scanned with CamScanner

Difference between Circuit Switching and Packet Switching

Circuit Switching	Packet Switching
In circuit switching there are 3 phases: i) Connection Establishment. ii) Data Transfer. iii) Connection Released.	In Packet switching directly data transfer takes place.
In circuit switching, each data unit know the entire path address which is provided by the source.	In Packet switching, each data unit just know the final destination address intermediate path is decided by the routers.
In Circuit switching, data is processed at source system only	In Packet switching, data is processed at all intermediate node including source system.

Scanned with CamScanner

Delay between data units in circuit switching is uniform.	Delay between data units in packet switching is not uniform.
Resource reservation is the feature of circuit switching because path is fixed for data transmission.	There is no resource reservation because bandwidth is shared among users.
Circuit switching is more reliable.	Packet switching is less reliable.
Wastage of resources are more in Circuit Switching	Less wastage of resources as compared to Circuit Switching
It is not a store and forward technique.	It is a store and forward technique.
Transmission of the data is done by the source.	Transmission of the data is done not only by the source, but also by the intermediate routers.

Scanned with CamScanner

Congestion can occur during the connection establishment phase, because there might be a case where a request is being made for a channel but the channel is already occupied.	Congestion can occur during data transfer phase, large number of packets comes in no time.
Circuit switching is not convenient for handling bilateral traffic.	Packet switching is suitable for handling bilateral traffic.
In Circuit switching, charge depend on time and distance, not on traffic in the network.	In Packet switching, charge is based on the number of bytes and connection time.
Recording of packet is never possible in circuit switching.	Recording of packet is possible in packet switching.
In Circuit Switching there is a physical path between the source and the destination	In Packet Switching there is no physical path between the source and the destination

Scanned with CamScanner

Circuit Switching does not support store and forward transmission

Packet Switching supports store and forward transmission

Scanned with CamScanner